

WEEK 01 PRACTICAL

Database System Engineering and
Distributed Backend Development

BookFlow Library Database

Course Code: 25CS1302E

Database: bookflow_db

Practical Record

1. Objective

Create the BookFlow library database with Books and Members tables, apply the required primary-key, NOT NULL, UNIQUE, AUTO_INCREMENT, and CHECK constraints, insert sample records, verify the data, and demonstrate constraint enforcement.

2. Database and Table Design

Database: bookflow_db

Table	Column	Definition / Constraint
books	book_id	INT, AUTO_INCREMENT, PRIMARY KEY
books	title	VARCHAR(255), NOT NULL
books	isbn	VARCHAR(13), NOT NULL, UNIQUE
books	published_year	INT, CHECK (published_year < 2027)
members	member_id	INT, AUTO_INCREMENT, PRIMARY KEY
members	full_name	VARCHAR(100), NOT NULL
members	email	VARCHAR(150), NOT NULL, UNIQUE

3. SQL Implementation

Database and table creation:

```
CREATE DATABASE IF NOT EXISTS bookflow_db;
USE bookflow_db;

CREATE TABLE books (
  book_id INT AUTO_INCREMENT PRIMARY KEY,
  title VARCHAR(255) NOT NULL,
  isbn VARCHAR(13) NOT NULL UNIQUE,
  published_year INT,
  CONSTRAINT chk_published_year CHECK (published_year < 2027)
);

CREATE TABLE members (
  member_id INT AUTO_INCREMENT PRIMARY KEY,
  full_name VARCHAR(100) NOT NULL,
  email VARCHAR(150) NOT NULL UNIQUE
);
```

Sample data insertion:

```
INSERT INTO books (title, isbn, published_year) VALUES
('The Alchemist', '9780061122415', 1988),
('Clean Code', '9780132350884', 2008),
('Atomic Habits', '9780735211292', 2018);

INSERT INTO members (full_name, email) VALUES
('Anil Kumar', 'anil.kumar@example.com'),
('Priya Sharma', 'priya.sharma@example.com'),
('Ravi Verma', 'ravi.verma@example.com');
```

4. Verification Queries

Table structure:

```
DESCRIBE books;  
DESCRIBE members;
```

Data:

```
SELECT * FROM books;  
SELECT * FROM members;
```

5. Constraint Tests

```
INSERT INTO books (title, isbn, published_year)  
VALUES ('Fake Copy', '9780061122415', 2000);
```

```
INSERT INTO books (title, isbn, published_year)  
VALUES (NULL, '9999999999999', 2010);
```

```
INSERT INTO books (title, isbn, published_year)  
VALUES ('Time Traveler', '8888888888888', 2030);
```

```
INSERT INTO members (full_name, email)  
VALUES ('Anil Clone', 'anil.kumar@example.com');
```

Output 1 — Books Table Structure

SQL File 3* x

Limit to 1000 rows

```
10  );
11  CREATE TABLE members (
12      member_id INT AUTO_INCREMENT PRIMARY KEY,
13      full_name VARCHAR(100) NOT NULL,
14      email VARCHAR(150) NOT NULL UNIQUE
15  );
16  DESCRIBE books;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Field	Type	Null	Key	Default	Extra
▶	book_id	int	NO	PRI	NULL	auto_increment
	title	varchar(255)	NO		NULL	
	isbn	varchar(13)	NO	UNI	NULL	
	published_year	int	YES		NULL	

Result 1 x

Output 2 — Members Table Structure

SQL File 3* x

Limit to 1000 rows

```
11 • CREATE TABLE members (  
12     member_id INT AUTO_INCREMENT PRIMARY KEY,  
13     full_name VARCHAR(100) NOT NULL,  
14     email VARCHAR(150) NOT NULL UNIQUE  
15 );  
16 • DESCRIBE books;  
17 • DESCRIBE members;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Field	Type	Null	Key	Default	Extra
member_id	int	NO	PRI	NULL	auto_increment
full_name	varchar(100)	NO		NULL	
email	varchar(150)	NO	UNI	NULL	

Result 2 | Result 3 x | Read Only

Result Grid
Form Editor

Output 3 — Books Data

SQL File 3*

Limit to 1000 rows

```
20 ('Clean Code', '9780132350884', 2008),
21 ('Atomic Habits', '9780735211292', 2018);
22 • INSERT INTO members (full_name, email) VALUES
23 ('Anil Kumar', 'anil.kumar@example.com'),
24 ('Priya Sharma', 'priya.sharma@example.com'),
25 ('Ravi Verma', 'ravi.verma@example.com');
26 • SELECT * FROM books;
```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	book_id	title	isbn	published_year
▶	1	The Alchemist	9780061122415	1988
	2	Clean Code	9780132350884	2008
	3	Atomic Habits	9780735211292	2018
*	NULL	NULL	NULL	NULL

Result Grid

Form Editor

Output 4 — Members Data

The screenshot shows a SQL IDE window titled "SQL File 3* x". The script editor contains the following SQL statements:

```
21 ('Atomic Habits', '9780735211292', 2018);
22 • INSERT INTO members (full_name, email) VALUES
23 ('Anil Kumar', 'anil.kumar@example.com'),
24 ('Priya Sharma', 'priya.sharma@example.com'),
25 ('Ravi Verma', 'ravi.verma@example.com');
26 • SELECT * FROM books;
27 • SELECT * FROM members;
```

The "Result Grid" tab is active, displaying the results of the last executed query (SELECT * FROM members). The grid shows three rows of data:

member_id	full_name	email
1	Anil Kumar	anil.kumar@example.com
2	Priya Sharma	priya.sharma@example.com
3	Ravi Verma	ravi.verma@example.com

The interface includes a toolbar with various icons, a "Limit to 1000 rows" dropdown, and a "Filter Rows" input field. The bottom status bar shows "Result 11", "Result 12", "books 13", and "members 14 x".

Output 5 — Constraint Test

The screenshot shows a SQL IDE interface with a query editor, a result grid, and an action output pane.

SQL Query:

```
24 ('Priya Sharma', 'priya.sharma@example.com'),
25 ('Ravi Verma', 'ravi.verma@example.com');
26 • SELECT * FROM books;
27 • SELECT * FROM members;
28 • INSERT INTO books (title, isbn, published_year)
29 VALUES ('Fake Copy', '9780061122415', 2000);
30
```

Result Grid:

member_id	full_name	email
1	Anil Kumar	anil.kumar@example.com
2	Priya Sharma	priya.sharma@example.com
3	Ravi Verma	ravi.verma@example.com
NULL	NULL	NULL

Action Output:

#	Time	Action	Message	Duration / Fetch
85	09:19:37	DESCRIBE members	3 row(s) returned	0.000 sec / 0.000 sec
86	09:19:37	INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '9780061122415', 2000);	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
87	09:19:37	INSERT INTO members (full_name, email) VALUES ('Anil Kumar', 'anil.kumar@example.com');	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
88	09:19:37	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
89	09:19:37	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
90	09:19:37	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9780061122415', 2000);	Error Code: 1062. Duplicate entry '9780061122415' for key 'books.isbn'	0.015 sec

SQL Additions:

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Output 6 — Constraint Test

The screenshot shows a SQL IDE interface with a query editor at the top and a results/output pane at the bottom. The query editor contains the following SQL statements:

```
25 ('Ravi Verma', 'ravi.verma@example.com');
26 • SELECT * FROM books;
27 • SELECT * FROM members;
28 • INSERT INTO books (title, isbn, published_year)
29 VALUES ('Fake Copy', '9780061122415', 2000);
30 • INSERT INTO books (title, isbn, published_year)
31 VALUES (NULL, '999999999999', 2010);
```

Below the query editor, the 'Result Grid' shows the contents of the 'members' table:

member_id	full_name	email
1	Anil Kumar	anil.kumar@example.com
2	Priya Sharma	priya.sharma@example.com
3	Ravi Verma	ravi.verma@example.com
*	NULL	NULL

The bottom pane shows the 'Action Output' log, which details the execution of the queries and the resulting error:

#	Time	Action	Message
✓ 97	09:20:41	DESCRIBE members	3 row(s) returned
✓ 98	09:20:41	INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '9780061122415', 2000)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 99	09:20:41	INSERT INTO members (full_name, email) VALUES ('Anil Kumar', 'anil.kumar@example.com')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 100	09:20:41	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned
✓ 101	09:20:41	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned
✗ 102	09:20:41	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9780061122415', 2000)	Error Code: 1062. Duplicate entry '9780061122415' for key 'books.isbn'

Output 7 — Constraint Test

SQL File 3*

Limit to 1000 rows

```

27 • SELECT * FROM members;
28 • INSERT INTO books (title, isbn, published_year)
29   VALUES ('Fake Copy', '9780061122415', 2000);
30 • INSERT INTO books (title, isbn, published_year)
31   VALUES (NULL, '9999999999999', 2010);
32 • INSERT INTO books (title, isbn, published_year)
33   VALUES ('Time Traveler', '888888888888', 2030);

```

Result Grid

Filter Rows:

Edit

Export/Import

Wrap Cell Content

	member_id	full_name	email
▶	1	Anil Kumar	anil.kumar@example.com
	2	Priya Sharma	priya.sharma@example.com
	3	Ravi Verma	ravi.verma@example.com
*	NULL	NULL	NULL

Result Grid

Form Editor

Result 23

Result 24

books 25

members 26 ×

Apply

Output

Action Output

#	Time	Action	Message
✓ 109	09:21:10	DESCRIBE members	3 row(s) returned
✓ 110	09:21:10	INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '9780061122415', 2000);	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 111	09:21:10	INSERT INTO members (full_name, email) VALUES ('Anil Kumar', 'anil.kumar@example.com');	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 112	09:21:10	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned
✓ 113	09:21:10	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned
✗ 114	09:21:10	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9780061122415', 2000);	Error Code: 1062. Duplicate entry '9780061122415' for key 'books.isbn'

Output 8 — Constraint Test

SQL File 3*

Limit to 1000 rows

```

29 VALUES ('Fake Copy', '9780061122415', 2000);
30 • INSERT INTO books (title, isbn, published_year)
31 VALUES (NULL, '9999999999999', 2010);
32 • INSERT INTO books (title, isbn, published_year)
33 VALUES ('Time Traveler', '8888888888888', 2030);
34 • INSERT INTO members (full_name, email)
35 VALUES ('Anil Clone', 'anil.kumar@example.com');

```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	member_id	full_name	email
▶	1	Anil Kumar	anil.kumar@example.com
	2	Priya Sharma	priya.sharma@example.com
	3	Ravi Verma	ravi.verma@example.com
*	NULL	NULL	NULL

Result 27

Result 28

books 29

members 30

Apply

Output

Action Output

#	Time	Action	Message
✓ 121	09:21:47	DESCRIBE members	3 row(s) returned
✓ 122	09:21:47	INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '9780061122415', 2000);	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 123	09:21:47	INSERT INTO members (full_name, email) VALUES ('Anil Kumar', 'anil.kumar@example.com');	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 124	09:21:47	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned
✓ 125	09:21:47	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned
✗ 126	09:21:47	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9780061122415', 2010);	Error Code: 1062. Duplicate entry '9780061122415' for key 'books.isbn'

6. Conclusion

The BookFlow database was implemented with the required tables, sample data, verification queries, and constraint tests using MySQL Workbench.